

INDIAN INSTITUTE OF TECHNOLOGY GANDHINAGAR (IIT Gandhinagar)



Assignment Report

M.Tech in Integrated Circuit Design and Technology (ICDT)

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Subject Name:	PMIC
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Project 1: Design of an Analog LDO Regulator in SCL 180nm CMOS Technology

Checkpoint

- Design LDO with ideal opamp of 80 dB gain (Hint: Use VCVS) as error amplifier and pass transistor.
- Create testbench for PSRR, Transient analysis, Loop Gain and phase, line and load regulation.
- **Bonus [1 pt]:** LDO with 5T OTA as error amplifier and pmos as pass element (Working stable LDO, may or may not match all the specs).

Specifications

- **Input Voltage (V_{in}):** 1.8V
- **Output Voltage (V_{out}):** 1.5V
- **Load Current (I_{load}):** 50 mA (Nominal), 20 mA (light-load), 100 mA (full-load)
- **Output Capacitor (C_{out}):** 500 pF (OFF-CHIP)
- **Efficiency (η):** $\geq 83\%$ (across load range)
- **PSRR @ 100 kHz:** -40 dB
- **DC Error in V_{out} :** $\leq \pm 0.1$ mV
- **Load Regulation:** $\leq 2\%$ variation from light-load to full-load
- **Line Regulation:** $\leq 2\%$ for $\pm 2.5\%$ variation in input voltage
- **Transient Response:** ≤ 150 mV overshoot/undershoot for a 20mA to 50mA load step (both up and down with edge time of 10ns shown in the same plot) with settling time (within $\pm 2\%$ of regulated value) less than 125 ns.

Compensation Methods

- **Method 1:** C_m (ON-CHIP) ≤ 15 pF (should be minimized) with nulling resistor in series with Miller capacitor.
- **Method 2:** Buffer to be placed in between output of opamp and pass transistor.

Design Constraints & Notes

- Overdrive voltage of all transistors should be $> 0.15\text{V}$ (nominally).
- Use SCL 180nm PDK. Only one ideal voltage source at 1.8V (V_{in}) can be used.
- Design Bandgap reference (BGR) for V_{ref} and current reference for biasing the error amplifier.
- Performance specs (except transient and Load Reg) should be reported at nominal load current.
- Design must match specs at 0°C , 27°C (nominal), and 60°C .

Schematics of 5T-OTA, BGR and LDO

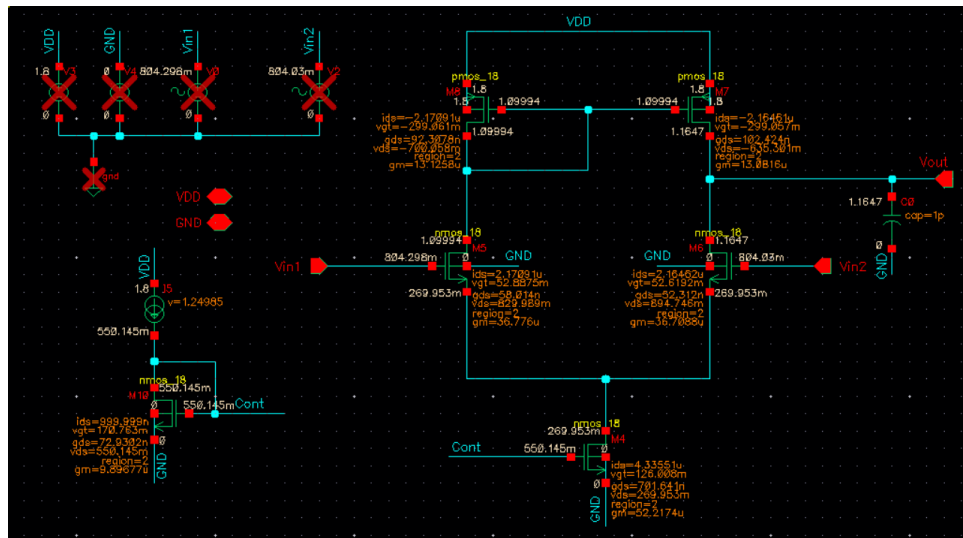


Figure 1: OTA DC parameters

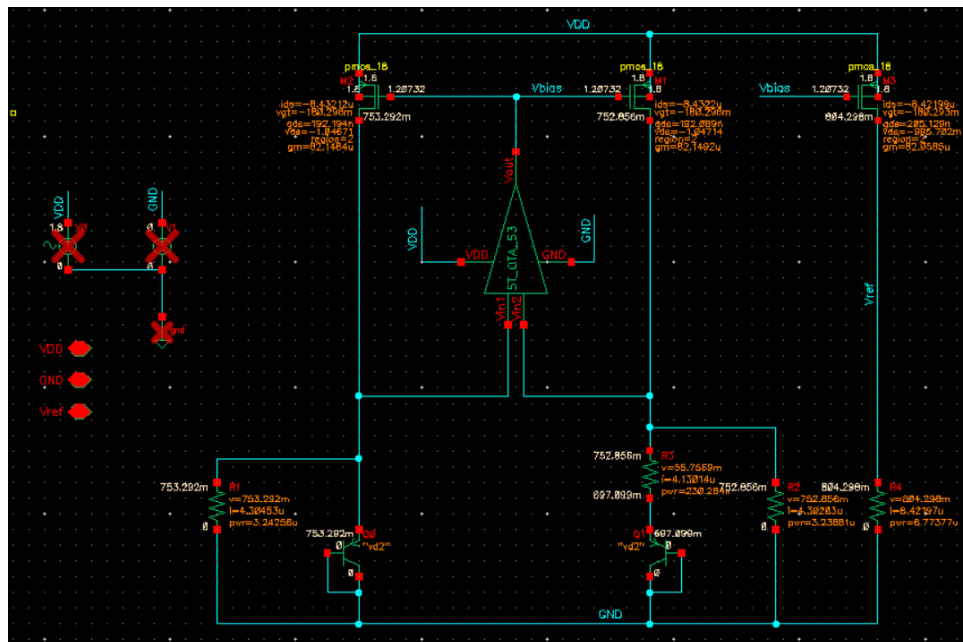


Figure 2: BGR DC Parameters

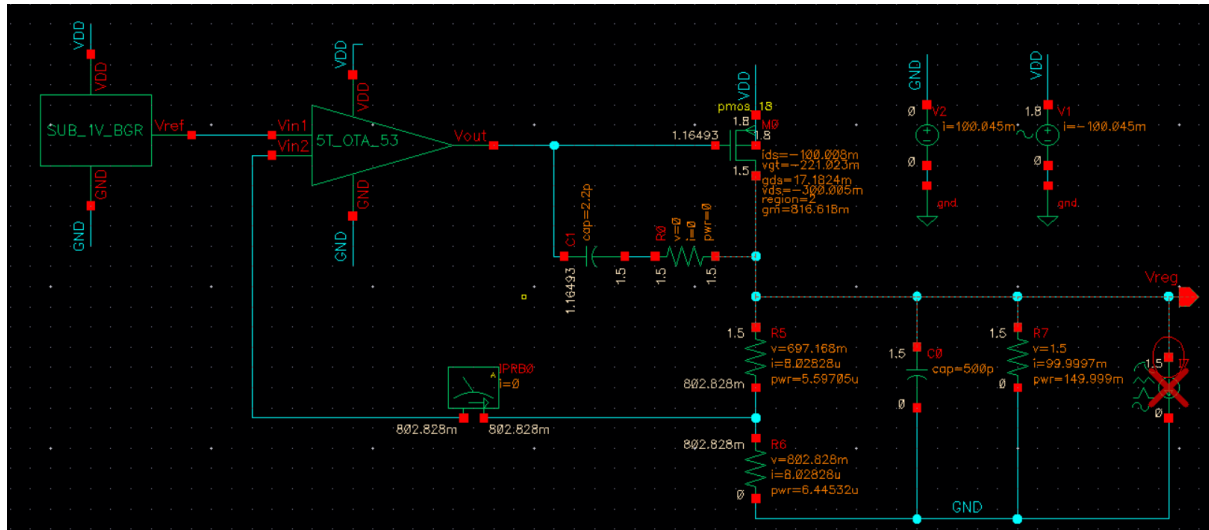


Figure 3: LDO DC parameters

Component Sizing Tables

Table 1: OTA Transistor Dimensions

Transistor	Type	Width (W)	Length (L)	Multiplier (m)
M4 (Tail)	NMOS	$1.7 \mu\text{m}$	$1.0 \mu\text{m}$	1
M5, M6 (Input)	NMOS	$2.2 \mu\text{m}$	$1.0 \mu\text{m}$	1
M7, M8 (Load)	PMOS	960 nm	$1.0 \mu\text{m}$	1
M10 (Bias)	NMOS	230 nm	$1.0 \mu\text{m}$	1

Table 2: BGR Transistor Dimensions

Transistor	Type	Width (W)	Length (L)	Multiplier (m)
M1, M2, M3	PMOS	$9.0 \mu\text{m}$	$1.0 \mu\text{m}$	1

Table 3: LDO Pass Transistor Dimensions

Transistor	Type	Width (W)	Length (L)	Multiplier (m)
M0 (Pass)	PMOS	$76.925 \mu\text{m}$	$1.0 \mu\text{m}$	1000

Simulation Plots

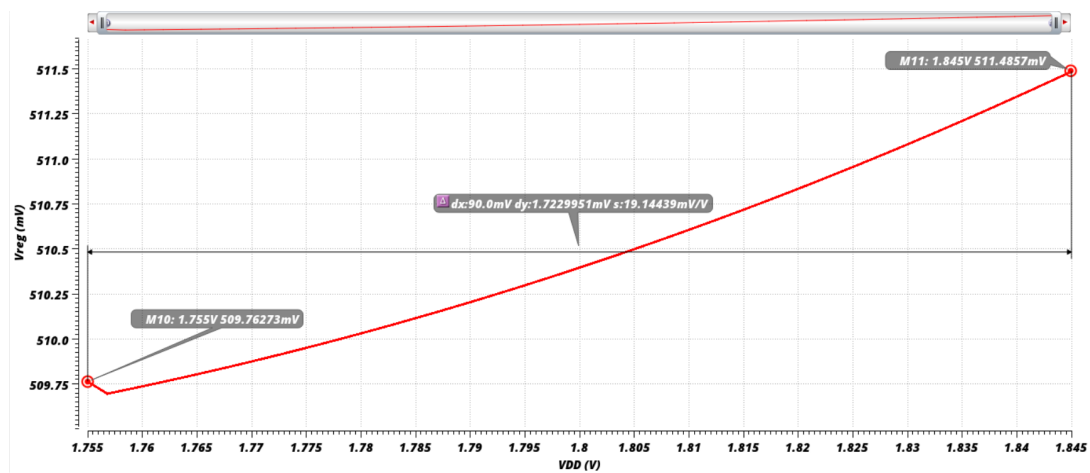


Figure 4: Line Regulation without Compensation

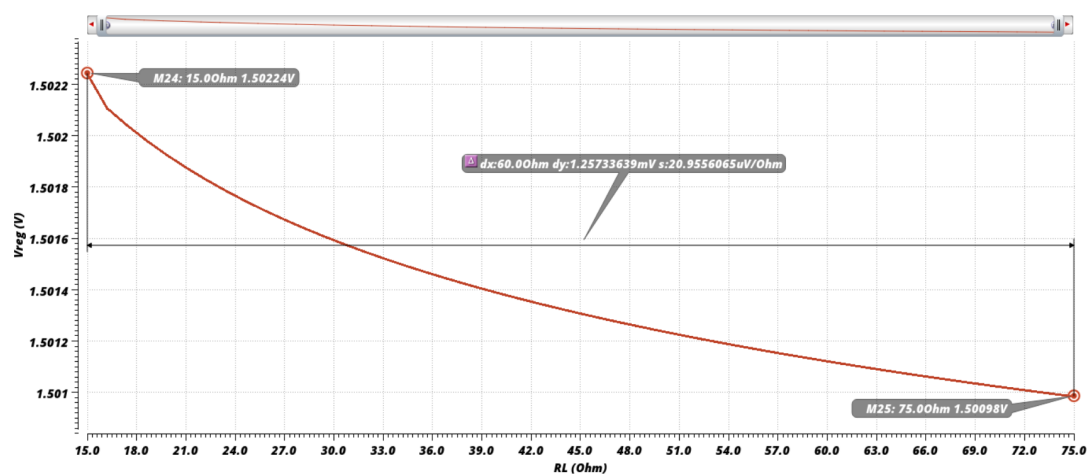


Figure 5: Load Regulation without Compensation

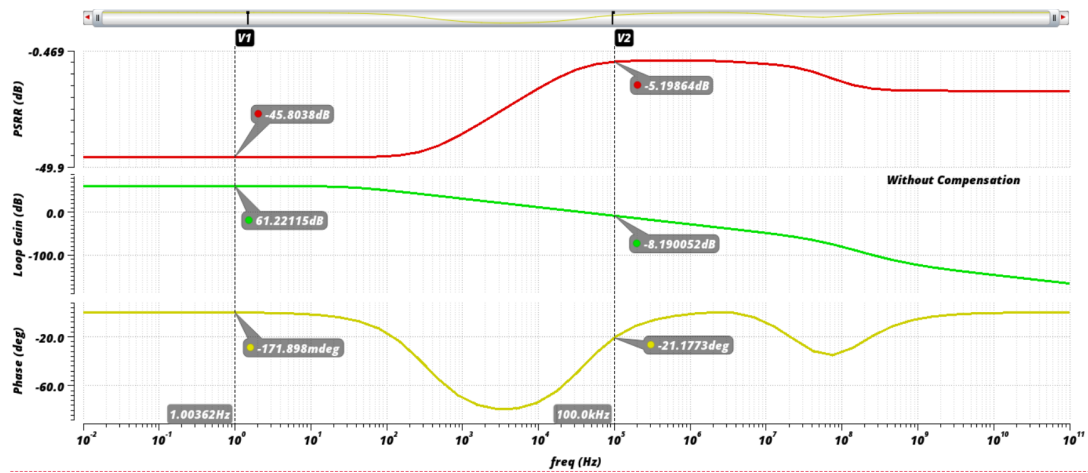


Figure 6: PSRR and Loop Gain without Compensation

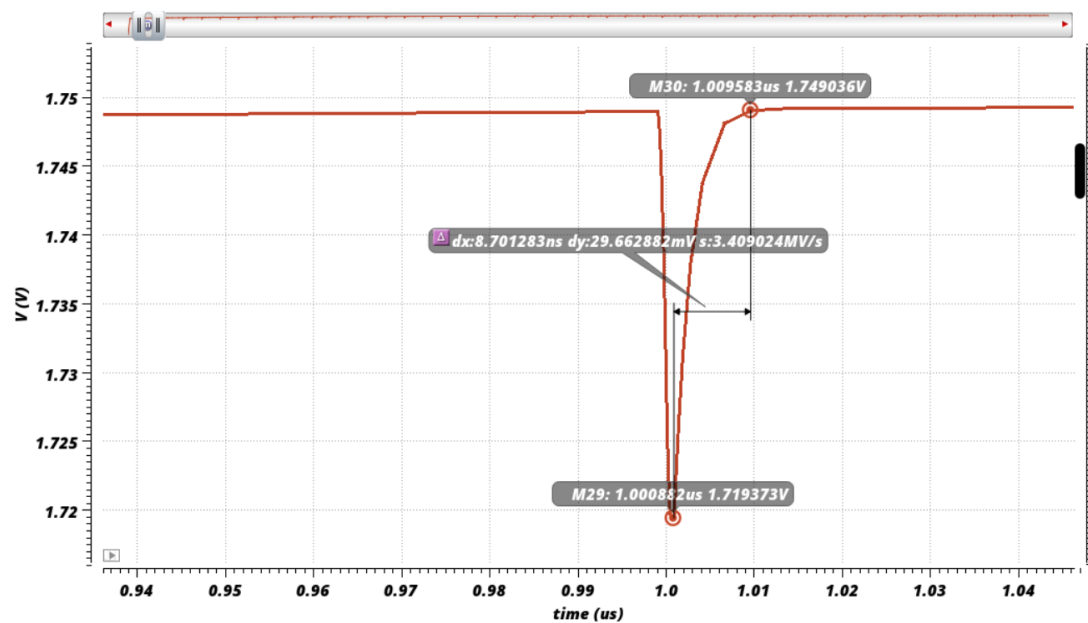


Figure 7: Undershoot with Compensation from 50mA to 20mA

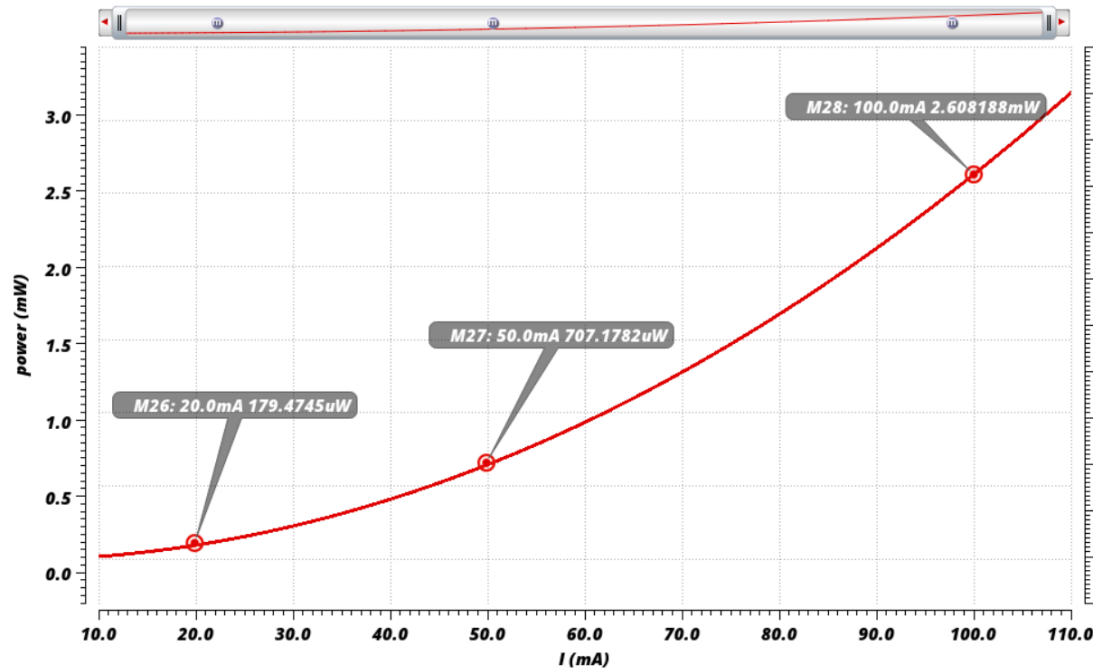


Figure 8: Power Consumption at different Load Currents

Simulation Results

Table 1: Amplifier Performance Parameters

Parameter	Simulated Value
DC Differential Gain	45.62 dB
Unit Gain Bandwidth (UGB)	3.31 MHz
Phase Margin	91°
Common Mode Gain	-31.68 dB
Total Power Dissipation	55.95 μ W
Peak-to-Peak Output Swing	801.14 mV

Table 2: LDO Regulator Performance Parameters

Parameter	Simulated Value
Loop Gain (DC)	61.22 dB
PSRR @ 0 Hz (DC)	-45.80 dB
PSRR @ 100 kHz	-5.19 dB
Line Regulation	19.14 mV/V
Load Regulation	20.95 μ V/ Ω
Nominal Power (50 mA)	707.18 μ W
Full-Load Power (100 mA)	2.61 mW

Conclusion and Discussion of Results

The LDO regulator design successfully achieved a DC Loop Gain of 61.22 dB and a DC PSRR of -45.80 dB. However, several target specifications were not met due to the following design constraints:

The PSRR at 100 kHz was measured at -5.19 dB, failing the -40 dB target. Due to the absence of optimized Miller compensation, which is necessary to maintain loop gain at higher frequencies.